Google Colab Guide - v2

Version 2. Current interface, screenshots captured 2026-11-10. Google changes Colab often, so buttons may move or be renamed. This guide is self-contained and does not depend on any workshop files.

1. What is Google Colab?

- Colab is a free, browser-based notebook environment. A notebook mixes code cells, their output, and text cells for notes. If you have used Jupyter notebooks, it works the same way, but runs in the cloud.
- Nothing to install: Python and many common data-science libraries are already available.
- Notebooks are saved in your Google Drive and have the extension .ipynb.
- The Runtime menu offers a free CPU, and GPU/TPU options when available (see section 11). Free capacity is limited and can change.
- A Gemini AI assistant is built in for eligible users (see section 12). Everything in this guide can be done without it.

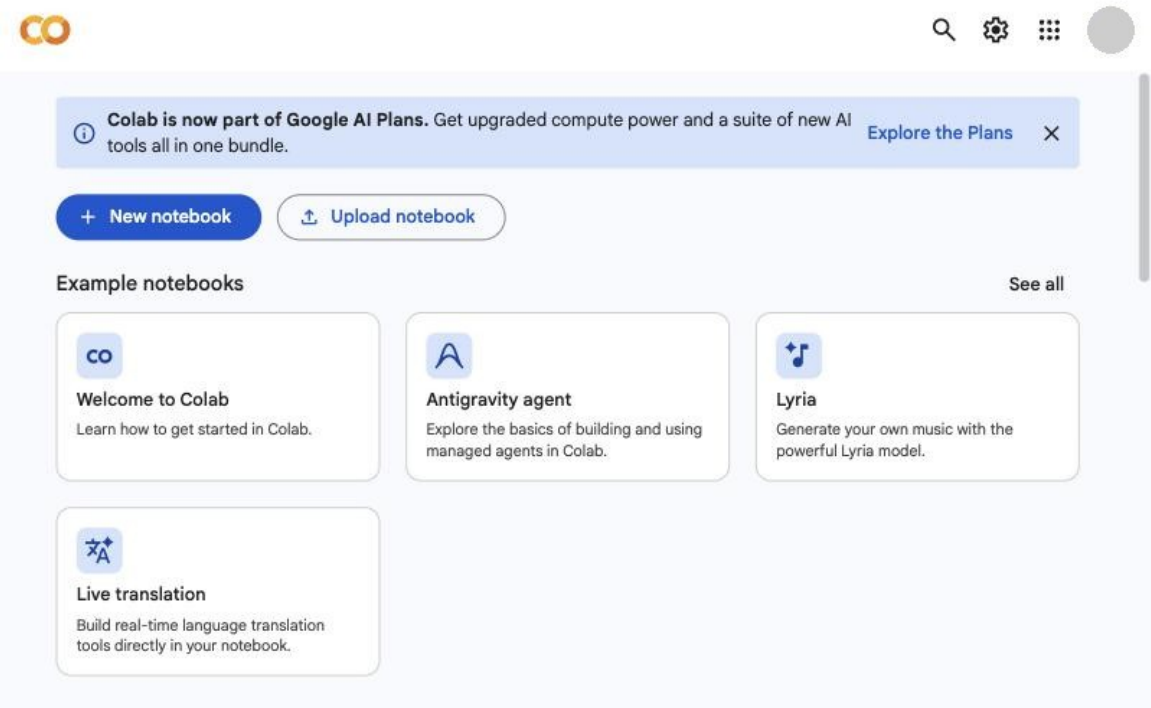
2. What you need

- A Google account (Gmail or a Google account created with another email). Create one at accounts.google.com/signup if you do not have one.
- A current web browser (Chrome, Edge, Firefox or Safari) and an internet connection.
- No payment or paid plan is needed for anything in this guide.

3. Open Colab

1. Go to colab.research.google.com and sign in if asked.

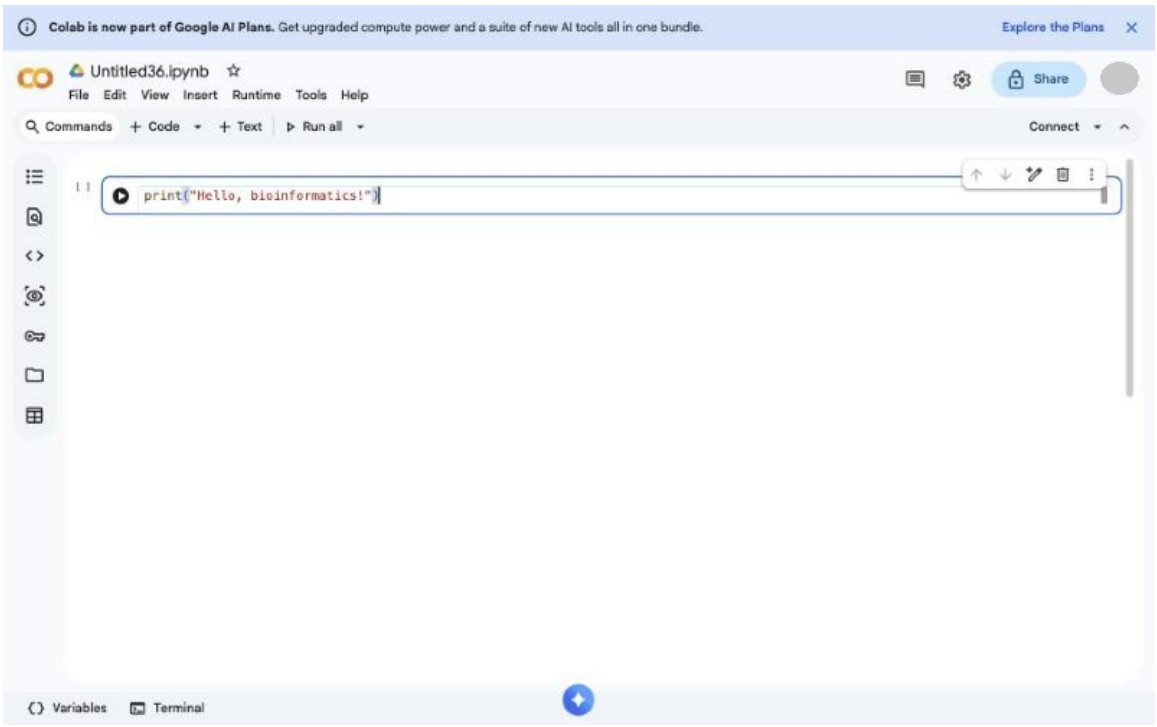
2. On the home page, click New notebook to start a blank notebook, Upload notebook to open a file from your computer, or choose an example notebook.



Colab home page. Your own recent notebooks are listed further down the page.

4. Create a new notebook

Click New notebook on the home page, or use File > New notebook in Drive from inside an existing notebook. The new notebook opens with one empty code cell. A Gemini side panel may open on the right; close it with the X if you do not want it.



A new notebook. The code cell in this screenshot already contains the example line from section 5, which was typed before the screenshot was taken; your new notebook starts with an empty cell.

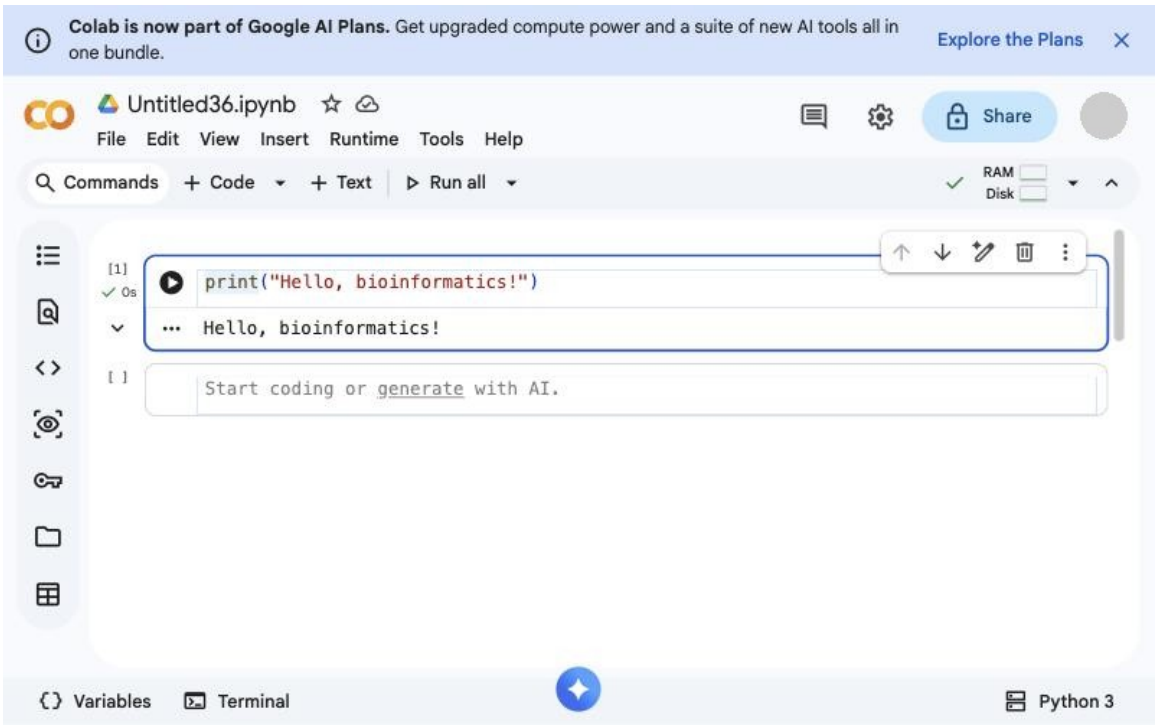
5. Write and run code

1. Click in the code cell and type a line of code (the example below).

2. Run the cell. Any of these works: click the round play button; press Cmd/Ctrl+Enter to run and stay in the same cell; press Shift+Enter to run and move to the next cell (Colab adds a new empty cell if there is none); press Alt+Enter (Option+Enter on a Mac) to run and always insert a new cell below. The first run can take a few seconds while Colab connects to a runtime.

3. The output appears under the cell and a green check mark shows it finished.

```
print("Hello, bioinformatics!")
```

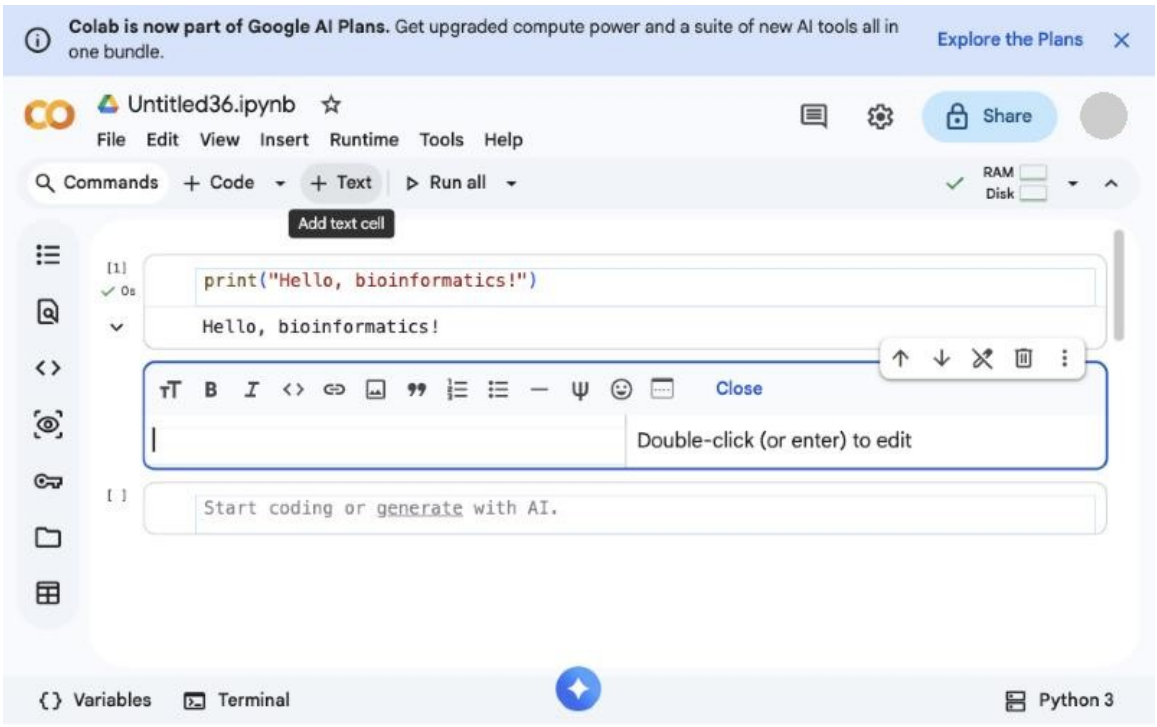


After running the cell: output below, green check mark, and an empty cell below (added here by Shift+Enter; the play button alone would not add a cell).

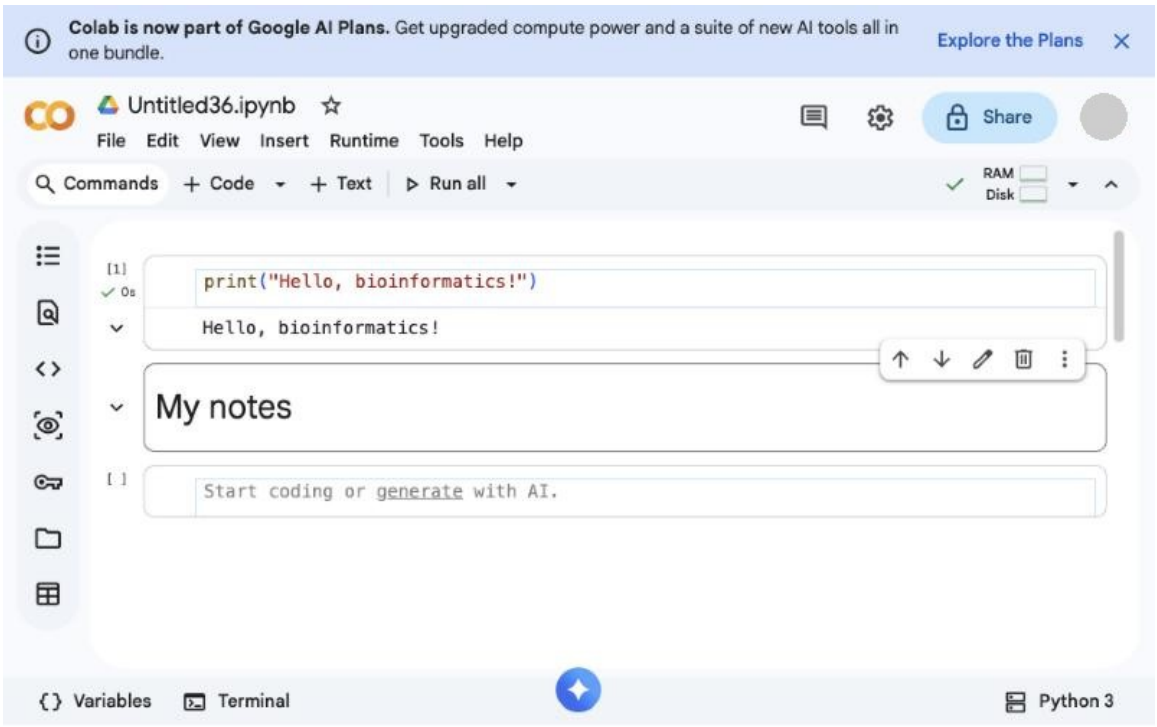
Shortcuts: Cmd/Ctrl+Enter runs the cell and stays; Shift+Enter runs the cell and selects the next one; Alt/Option+Enter runs the cell and inserts a new one below; Cmd/Ctrl+F9 runs all cells.

6. Add notes with text cells

Click + Text in the toolbar. Type in Markdown, for example "# My notes" for a heading. Click Close (or press Shift+Enter) to show the formatted text, and double-click the cell to edit it again.



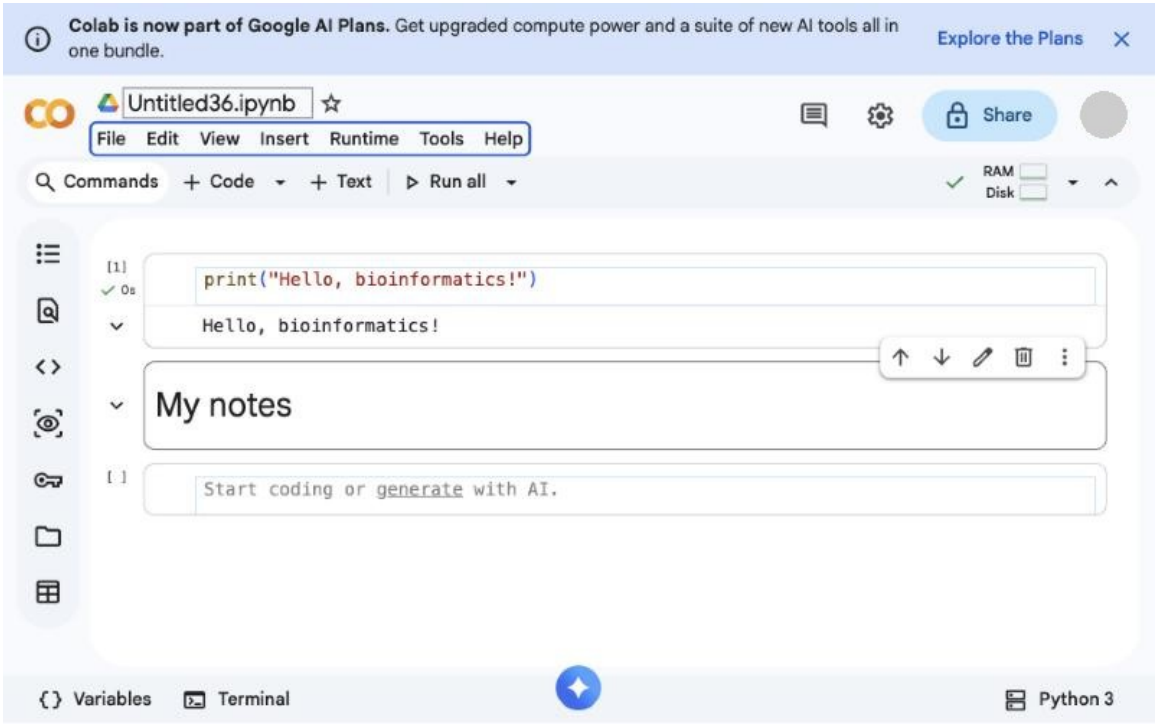
A new text cell in edit mode. It is still empty here; type your Markdown (for example # My notes) in the left half and a preview appears on the right.



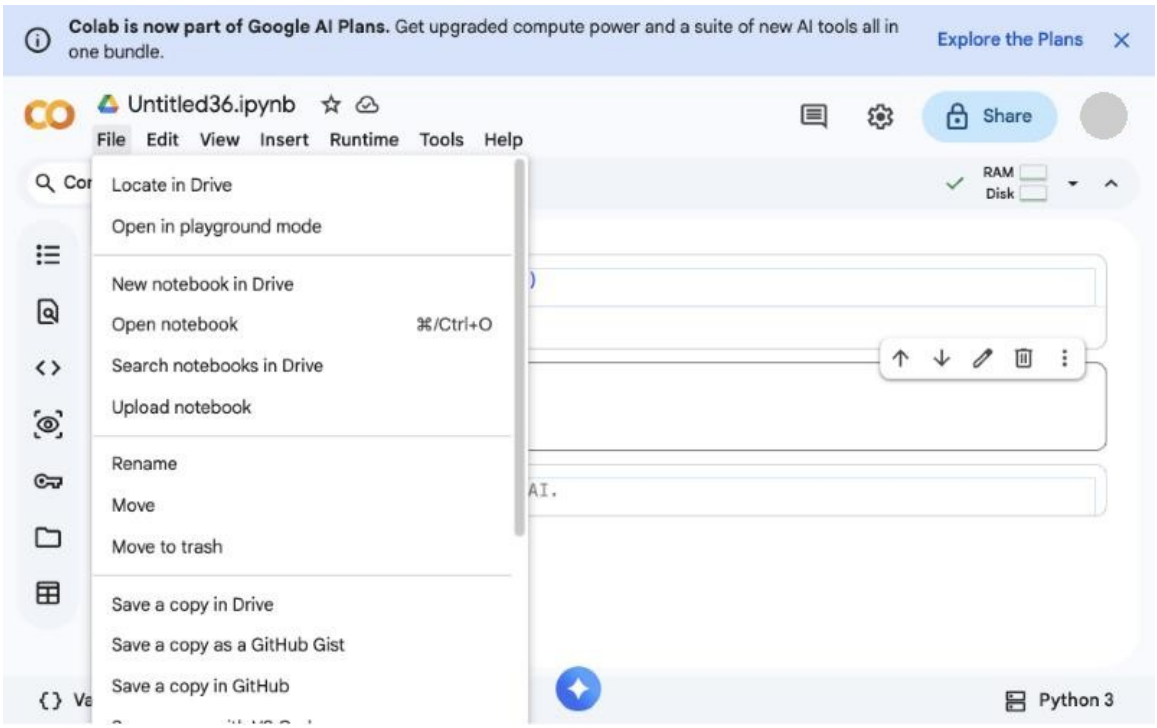
A text cell after closing the editor: the Markdown is shown as formatted text.

7. Name and save your notebook

- New notebooks are called Untitled0.ipynb, Untitled1.ipynb and so on. Click the name at the top left, type a new name and press Enter.
- Colab saves your notebook (its code, text and saved outputs) to your Google Drive automatically. Use File > Locate in Drive to find the file.
- File > Save a copy in Drive creates your own copy of a notebook that someone shared with you. Do this before editing a shared notebook.
- File > Move to trash removes a notebook from Drive (you can restore it from the Drive trash).



Click the notebook name to rename it.



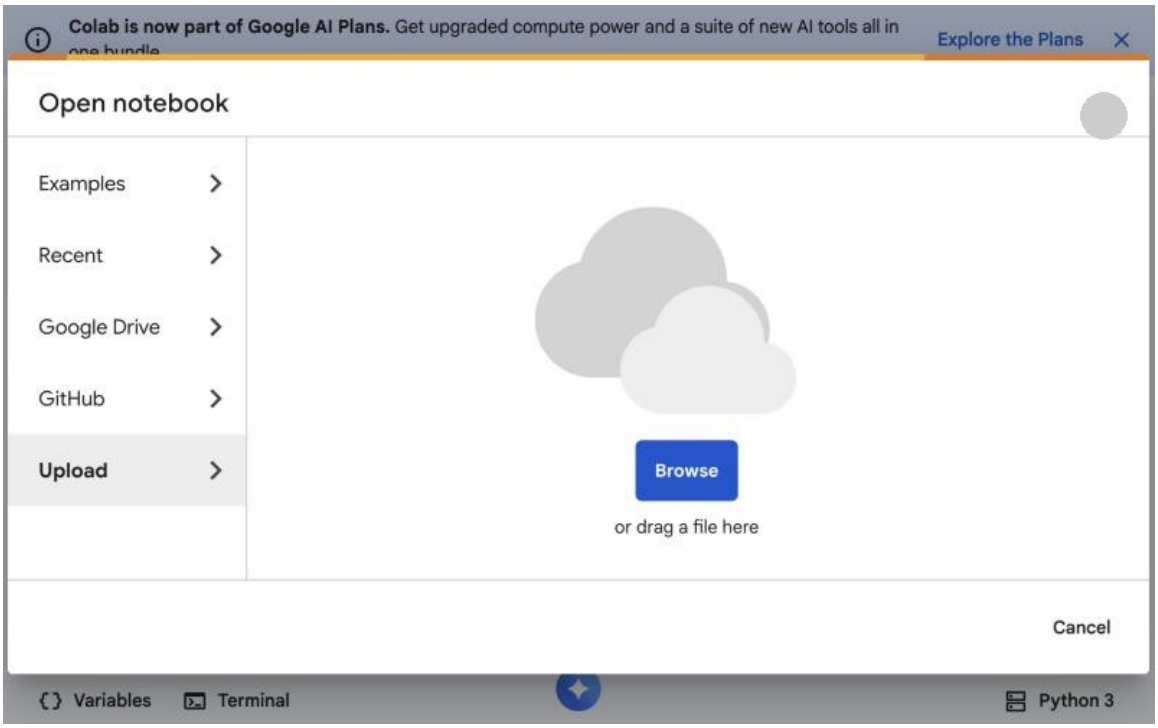
The File menu: Locate in Drive, New notebook in Drive, Open notebook, Upload notebook, Rename, Save a copy in Drive and more.

8. Open or upload an existing notebook

1. Choose File > Open notebook (Cmd/Ctrl+O) or File > Upload notebook.

2. Pick a tab: Examples, Recent, Google Drive, GitHub or Upload.

3. On the Upload tab, click Browse, or drag an .ipynb file into the box. The notebook opens in a new tab.



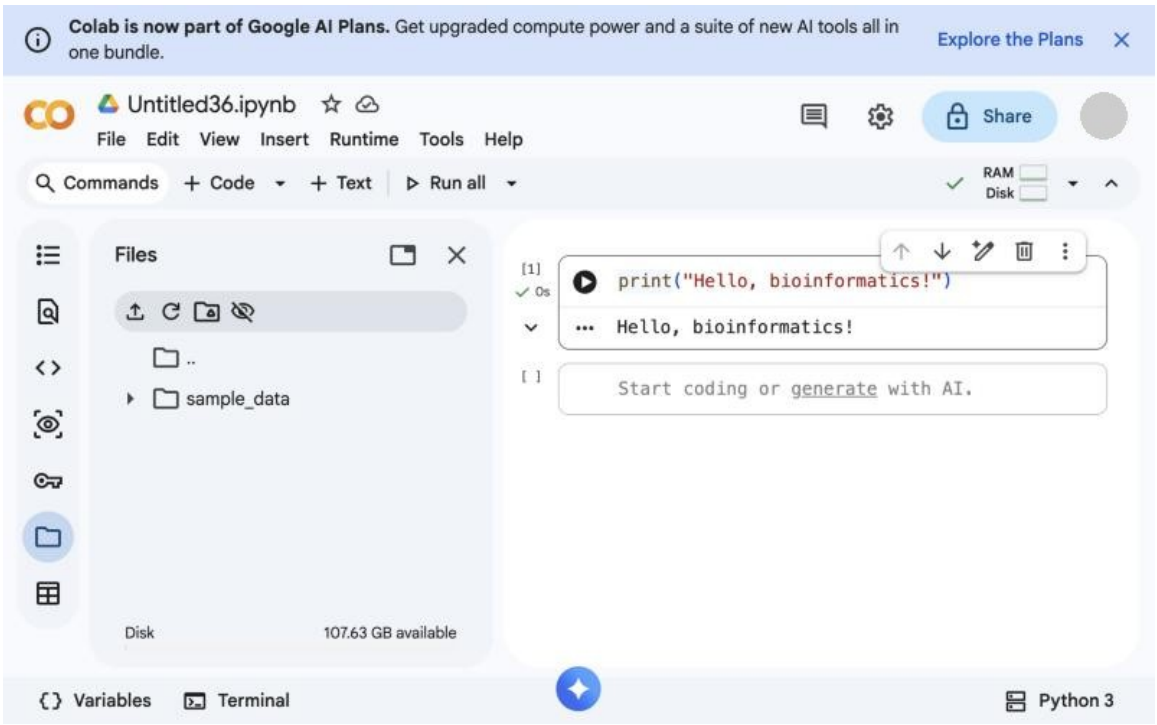
The Open notebook dialog, Upload tab. The other tabs list notebooks from Drive and GitHub.

Only open notebooks from sources you trust, because a notebook can run code on your Colab runtime and read files you connect to it.

9. Upload data files and know what is saved where

Click the folder icon in the left toolbar to open the Files pane. It shows the temporary file space of your current runtime (the virtual machine, or VM, that Google lends you), including a sample_data folder you can ignore or delete.

1. Click the upload icon at the top of the Files pane (or drag a file into the pane).
2. Read it in code, for example with pandas.read_csv("my_file.csv").
3. Download or copy results you want to keep before the VM is lost (see below).



The Files pane. The icons at the top are Upload, Refresh, Mount Drive (the folder with a Drive symbol) and Hide hidden files. Mount Drive connects your Google Drive; it does not create a new folder.

Four different things can be saved or lost, and they behave differently:

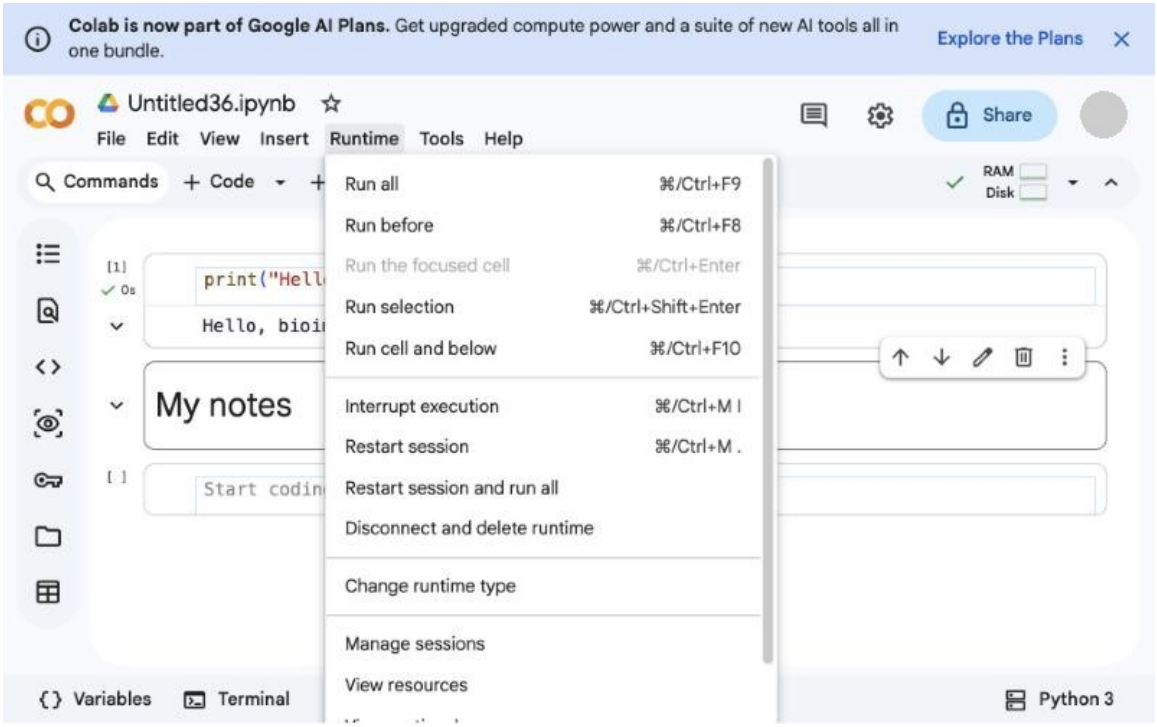
- The notebook itself (code, text and saved outputs) is autosaved to Drive.
- Variables and loaded data live in memory. They are cleared when you restart the session or the VM is lost.
- Files in the Files pane (the /content folder) and packages you installed with !pip live on the VM's disk. A simple restart of the session normally keeps them, but they are deleted when the VM is deleted, replaced or times out.
- Files you explicitly write to a mounted Google Drive folder (for example /content/drive/MyDrive/) persist. Mounting Drive alone does not back up /content; you must save or copy files there, or download them.

To keep files between sessions you can connect Google Drive (use the Drive icon in the Files pane or a short mount snippet). Only do this when a notebook needs it, because it gives the notebook access to your Drive.

10. The Runtime menu

- Run all, Run before, Run the focused cell and Run cell and below: run groups of cells. Run selection is different: it runs only the code you have highlighted inside a cell.
- Interrupt execution stops code that is running. Restart session clears variables and starts a fresh session on the same VM; installed packages and files on the disk are usually kept. Disconnect and delete runtime releases the VM, and everything on its disk is lost.

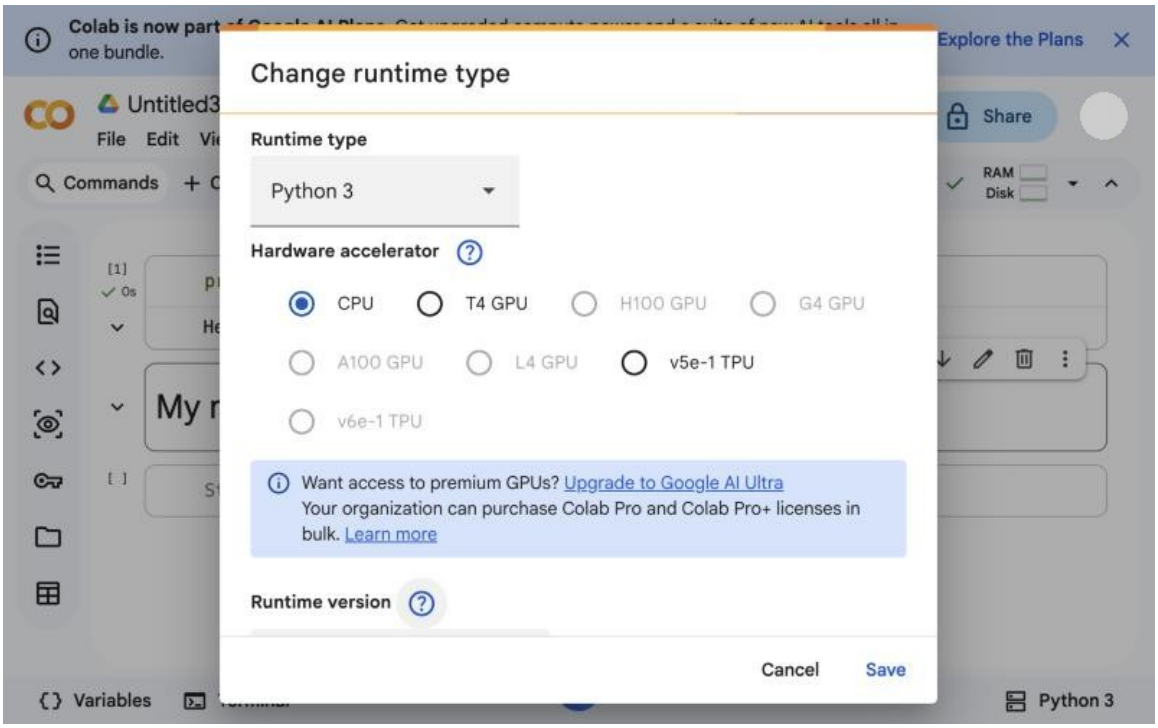
- Change runtime type (next section) selects Python and the hardware.
- Manage sessions and View resources show what is running and how much RAM and disk you use.



The Runtime menu.

11. Choose hardware (CPU, GPU, TPU)

Open Runtime > Change runtime type. The default is Python 3 on CPU, which is enough for most beginner work. A T4 GPU or a TPU can be selected when available; some options (for example H100 or A100) are greyed out on the free tier and need a paid plan. Free GPU time is limited and can be unavailable at busy times; paid plans do not guarantee capacity either. Choose Save to apply the change; it may restart your runtime.



Change runtime type dialog.

12. Gemini assistant in Colab

For eligible users (Google currently requires you to be at least 18 and in a supported location), the blue sparkle button at the bottom of the notebook toggles a Gemini chat panel, and empty code cells offer a "generate with AI" link. You can ask it to explain code, fix errors or draft a snippet. If it is not available to you, you can do everything in this guide without it.

Always read and test generated code. The assistant can use the code and outputs in your notebook as context, and Google may keep and have people review AI interactions (see the Colab FAQ for the current retention period). So do not put confidential, patient, restricted or unpublished data in prompts, in the notebook, or in cell outputs when you use it.

13. Secrets and API keys

The key icon in the left toolbar opens Secrets, where you can store keys and tokens privately. Saving a secret is not enough: you must switch on Notebook access for each notebook that may use it, and your code must read it, for example with the lines below. Granting access lets the code running in that notebook read the key, so only grant it to notebooks you wrote or have reviewed. Never print the value, never type a key directly into a code or text cell, and never share a notebook without checking it.

```
from google.colab import userdata
key = userdata.get("MY_SECRET_NAME")   # do not print(key)
```

Note that Secrets do not create environment variables, so code that reads os.environ will not find them unless you copy the value there yourself.

14. How Colab differs from Jupyter

- Everything runs on a temporary Google machine, not on your computer. Files on that machine are lost when the VM is deleted, replaced or times out (see section 9 for what a simple restart does and does not remove).
- Colab may collapse groups of cells under headings; click the arrow next to a section to expand it.
- Some HTML and CSS styling in text cells does not render.
- Packages that are not preinstalled can be added in a cell with: !pip install package_name (this lasts until the VM is lost).
- Idle sessions disconnect after a while; free sessions also have a maximum length.

15. Troubleshooting

- Cell will not run or stays on "Connecting": check your internet, then use Runtime > Restart session.
- "Google sign-in required": sign in with a Google account; some school accounts block Colab, so ask your IT team or use a personal account.
- You cannot edit a shared notebook: use File > Save a copy in Drive.
- Out of memory or a crash: Runtime > Restart session, and load smaller data.
- No GPU available: use CPU, or try again later.
- Lost your notebook: File > Open notebook > Recent, or search your Google Drive.

16. Data and account safety

- Use only data you are allowed to share, preferably public or synthetic.
- Do not upload identifiable, clinical, confidential or unpublished data unless your institution has approved it.
- Saved outputs are part of the notebook. Before you share a notebook, clear or review its outputs (Edit > Clear all outputs) so it does not expose data, file listings or printed secrets.
- When you use Gemini, keep restricted data out of the notebook, its outputs and the prompts (see section 12).
- Mount Google Drive, grant Secrets access and share notebooks only when you understand the access you are granting.
- Review notebooks from other people before running them.